

Feature Engineering Report

Forecasting Botswana Monthly Food Price Inflation | IndabaX Botswana 2026 Hackathon
Phase 1 Submission — condensed report (4-page limit)

1. Problem Framing and Target Variable

This project forecasts Botswana monthly food price inflation. The target variable is **FAO_23014** (FAO Botswana consumer prices, Item Code 23014: food price inflation). Rows with a missing target were dropped, giving a final modelling table of **276 monthly observations from 2001-01 to 2023-12** with no missing values, no duplicate rows, and no duplicate months. Because all predictor series end in 2023-12, the forecast horizon has no contemporaneous features; Section 8 documents the strategy for missing future predictors, and no post-2023 feature values are used anywhere in training or evaluation.

Food inflation in Botswana is driven through several channels that motivate the multi-dataset design: Botswana is landlocked and imports the majority of its food, largely from or through South Africa, so global shipping costs reach domestic shelves through landed import costs at regional ports; fuel prices transmit through long inland road-freight distances to Gaborone and secondary towns; and the Bank of Botswana policy rate captures domestic demand and credit conditions. The feature set is built to represent each channel explicitly.

2. Raw Data Sources and Audit

Dataset	File	Frequency	Coverage	Rows / missing	Role
Baltic Dry Index (BDI)	01_baltic_dry_index_daily.csv	Daily	2000-01-04 to 2023-12-22	5,992 / 0	Global dry-bulk shipping cost proxy; engineered to monthly features
Brent crude oil	02_brent_crude_monthly.csv	Monthly	2000-01 to 2023-12	288 / 0	Global fuel and transport cost pressure
Botswana policy rate	03_botswana_policy_rate.csv	Monthly	2000-01 to 2023-12	288 / 0	Domestic monetary policy and demand conditions
FAO Botswana consumer prices	04_fao_botswana_prices.csv	Monthly	2000-01 to 2023-12	852 / 0	Target (23014) plus general CPI (23012) and food CPI (23013)
Human Capital Project (HCP)	05_human_capital_project.csv	Monthly	2000-01 to 2023-12	4,320 / 60	Cross-country indicators; used in the HCP linkage analysis, excluded from the main forecasting table (see Section 5)

Every raw file was audited for row count, coverage, missingness, duplicates, and chronological ordering before merging, so the modelling table was not built blindly. All artifacts use relative paths: the cleaned table is **Data_Clean/prediction_data.csv** and final deep-learning predictions are **Data_Clean/model_2_lstm_final_predictions.csv**, regenerated by **Data_Cleaning.ipynb** and **Model_2_LSTM.ipynb** respectively.

3. Daily-to-Monthly BDI Feature Engineering

The BDI is the only daily series, so it required explicit aggregation to the monthly target frequency. The raw file contains **Date**, **BDI_Close**, **BDI_High**, and **BDI_Low**. Close values were used as the basis for aggregation because the daily close is the settled index value and gives a stable foundation for volatility and return calculations. Each month was keyed as **year_month** (YYYY-MM) to align with all other monthly datasets.

Engineered variable	Method	What it captures	Transmission reasoning for Botswana
BDI_std	Standard deviation of daily BDI_Close within each month	Within-month volatility of shipping costs	Volatility signals supply-chain uncertainty and cost shocks. Because imported food and inputs are priced on contracts exposed to freight conditions, instability in shipping markets can precede domestic food price adjustments.
monthly_return_bdi	(last – first) / first, using the first and last daily close of the month	Direction and size of the shipping-cost move during the month	Sustained rises in dry-bulk freight raise the landed cost of grain and bulk food imports moving through regional ports into landlocked Botswana, with a lag as contracts and stock turn over.
first, last	First and last daily close per month	Intermediate values only	Retained in the cleaned CSV for auditability but excluded from model inputs, since they duplicate BDI level information already carried by the return.

Keeping both volatility and return gives the model two distinct views of shipping pressure: uncertainty (BDI_std) and movement (monthly_return_bdi). BDI_High and BDI_Low were inspected but not included; monthly range-style features (for example high–low range or coefficient of variation) are logged as candidates to be added only if they improve validation RMSE.

4. Other Engineered and Selected Variables

The FAO data arrives in long format (Date, Item Code, Item, Value) and was pivoted to wide format so each indicator becomes a column. All monthly series were merged on year_month, sorted chronologically, and filtered to rows where the target is observed.

Variable	Source / method	Justification
FAO_23012	General CPI, 2015=100 (pivoted)	Controls for broad domestic inflation pressure; food inflation shares macro shocks and cost drivers with headline CPI.
FAO_23013	Food CPI, 2015=100 (pivoted)	Underlying food price level; provides trend information closely related to the target.
Brent_USD_per_barrel	Monthly Brent price, merged on year_month	Fuel feeds directly into road-freight costs over Botswana's long inland distribution distances, and indirectly into production inputs such as fertiliser and logistics.
policy_rate	Bank of Botswana policy rate, merged on year_month	Captures monetary conditions: higher rates dampen demand-side pressure, while rate changes also respond to existing inflation trends.
month_sin, month_cos	Cyclical month-of-year encoding (Model 2 only)	Encodes agricultural and demand seasonality without a discontinuity between December and January.

5. Feature Selection Decisions

Features were selected deliberately, balancing predictive signal, economic interpretability, and the limited 276-month sample:

Decision	Choice and reason
Historical target as input	FAO_23014 history is used as a model input only up to the month before the forecast month. Food inflation is persistent, so correctly lagged target history is a valid and strong predictor.
HCP variables	Excluded from the main forecasting table: the file has missing values (60) and many indicators, and adding them without dedicated feature selection would add noise and reduce explainability. They are analysed separately in the HCP linkage memo.
BDI first / last	Excluded from model inputs (intermediate values; information duplicated by the monthly return).
No stored lag columns	The cleaned CSV keeps contemporaneous monthly variables only; each model builds its own lagged structure at modelling time. This keeps one base table usable by both the classical and deep-learning models and reduces accidental leakage from pre-built lag columns.

6. Lag Structure and Leakage Controls (Model 2, LSTM)

For a target month t , the LSTM input sequence ends at month $t-1$, making the task a genuine one-month-ahead forecast with no same-month information. Lookback windows of 6, 12, and 18 months were tested (short momentum, full annual seasonality, extended history), each with 16 and 32 hidden units, and the final configuration was selected on validation RMSE only. Features and target are standardised with statistics fitted on the training period only; validation and test rows are transformed with training-period parameters. Any future rolling statistics will be shifted so no window touches the target month.

Chronological split	Range	Purpose
Training	2001-01 to 2017-12	Fit scalars and train candidate models
Validation	2018-01 to 2020-12	Select final configuration
Test (untouched)	2021-01 to 2023-12	Final evaluation and prediction export (36 rows)

7. Baseline Comparison and Honest Evaluation

All methods are evaluated on the same untouched 2021-01 to 2023-12 target months, with no test leakage in scaling or model selection, so the comparison is fair by construction. Exported columns: date, actual, LSTM, persistence, seasonal_naive, residual, absolute_error.

Method	MAE	RMSE	Bias (actual – predicted)	Interpretation
Persistence baseline	1.4593	2.1571	0.0690	Previous month's inflation; strong short-term benchmark.
Seasonal naive baseline	6.3642	7.5079	2.7361	Same month last year; weak over the volatile test window.
Model 2 LSTM	3.6084	4.5113	2.8524	Beats seasonal naive but does not yet beat persistence.

We report this honestly: the LSTM has learned structure beyond annual repetition, but persistence remains hard to beat on this window. Planned improvements target bias reduction and short-term responsiveness (smaller models, stronger regularisation, differenced targets, or modelling residuals against persistence), plus a matching prediction export from the classical model on the same 36 months.

8. Forecast-Horizon Strategy for Missing Future Predictors

All predictors end at 2023-12, so forecasting beyond that point requires an explicit, documented strategy for future feature values — using realised future feature values would be leakage and is avoided throughout.

Approach	Description	Strength / risk
Forecast external predictors	Separately forecast Brent, BDI features, and the policy rate, then feed them in.	Uses the full feature set; predictor errors compound.

Approach	Description	Strength / risk
Scenario-based predictors	Low / medium / high paths for oil, shipping, and policy.	Strong for policy interpretation; not a single deterministic forecast.
Target-history-only model	Lagged food inflation and calendar seasonality only.	No unknown future inputs needed; misses external cost shocks.
Hybrid (selected)	Target-history features drive the main forecast; external variables are used for scenario sensitivity.	Practical within the timeline; assumptions stated explicitly.

9. Limitations and Conclusion

Limitations: HCP indicators are not yet in the forecasting table; BDI high/low range features and rolling mean/volatility features are logged but untested in the leakage-safe framework; and the LSTM currently trails the persistence baseline on the test window. The pipeline nonetheless delivers a clean, audited, reproducible monthly table integrating domestic prices, shipping, fuel, and monetary policy — with the key engineering contribution being that every variable is aligned to forecast timing: for any target month, models see only prior history. Run order: Data Cleaning.ipynb → Model_1_Statistical.ipynb → Model_2_LSTM.ipynb → compare all methods on the same target months.